Algorithmic Trading Strategy

UMN FMA Trading Competition

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What We Built

Two autonomous strategies trading US equities & ETFs

Strategy	**Allocation**	**Asset(s)**
LSTM Portfolio	85%	VTI, SCHZ, PDBC, VIXM
RSI-2 Mean Reversion	15%	QQQ

All trades execute daily before market close via Alpaca API

Strategy 1: LSTM Portfolio (85%)

A neural network decides how to split money across 4 asset classes **every day**

- **VTI**: US Stocks
- **SCHZ**: Bonds
- **PDBC**: Commodities
- **VIXM**: Volatility

Phase	Period	Purpose
Training	Jan 2016 to Feb 2025	Model learns patterns
Validation	Mar 2025 to Feb 2026	Prevents overfitting
Live (competition)	Apr 2026 onward	Truly unseen data

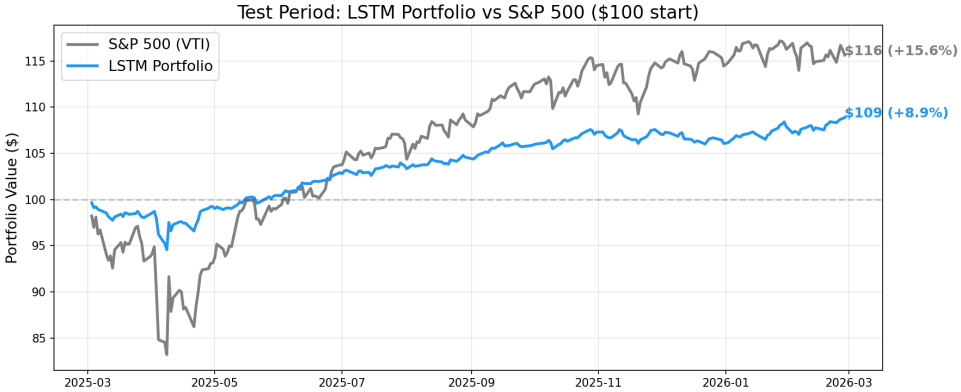
Optimized for **Sharpe ratio**: maximizes returns *per unit of risk*

How the LSTM Works

1. Feeds in last **50 days** of prices & returns for all 4 assets
2. LSTM processes the sequence, learns patterns
3. Outputs **4 weights** (e.g., 28% stocks, 13% bonds, 30% commodities, 29% volatility)
4. Rebalances portfolio to match

Key: In uncertain markets, the model *automatically* shifts toward bonds, commodities, and volatility

LSTM Backtest: Validation Period (Mar 2025 to Feb 2026)



S&P 500: +15.6% return
Max drawdown: −16%

Our LSTM: +8.9% return
Max drawdown: **only −5%**

Max drawdown = largest peak-to-trough decline during the period. Lower is better.

Strategy 2: RSI-2 Mean Reversion on QQQ (15%)

RSI (Relative Strength Index) measures how overbought or oversold an asset is

We use a 2-day RSI, computed from the last 2 daily price changes:

$$\text{RSI}(2) = 100 - \frac{100}{1 + \dfrac{\text{average gain over 2 days}}{\text{average loss over 2 days}}}$$

- RSI near 0 = extremely oversold (price dropped sharply)
- RSI near 100 = extremely overbought (price surged)

Our rules:

- **Buy QQQ** when $\text{RSI}(2) < 10$: sharp drop, expect a bounce
- **Sell QQQ** when $\text{RSI}(2) > 70$: let profits run, then take profit
- Typical hold: 1–5 days

RSI-2 Backtest: Performance During Market Crises

Period	**Dates**	**S&P 500**	**QQQ**	**RSI-2**
Russia-Ukraine War	01-Feb-22 to 30-Jun-22	−16.1%	−22.8%	**+2.0%**
2022 Bear Market	01-Jan-22 to 31-Dec-22	−19.5%	−33.0%	**+3.3%**
2025 Tariff Crisis	01-Mar-25 to 30-May-25	−0.8%	+2.2%	**+13.4%**
Recent 3 Months	01-Dec-25 to 27-Feb-26	+0.4%	−1.9%	**+1.9%**

Risk Management

1. **Diversified**: 4 asset classes, not just stocks
2. **AI-driven defense**: LSTM shifts to safe havens automatically
3. **Position sizing**: 85/15 split limits RSI exposure
4. **Long-only**: no leverage, no shorts, no options
5. **Daily rebalancing**: never holds stale positions
6. **All orders before close**: fills at real-time prices

Execution

Infrastructure

- Linux server running 24/7
- Python + PyTorch + Alpaca API
- 2 systemd services with auto-restart on failure

Daily Schedule (ET)

- 3:45 PM: LSTM fetches data, runs model, rebalances 4 ETFs
- 3:50 PM: RSI-2 checks QQQ signal, trades if triggered
- All orders fill before 4:00 PM close
- Max 4–5 orders per day

Summary

1. **LSTM (85%)**: AI allocates across stocks, bonds, commodities, volatility
2. **RSI-2 (15%)**: Buys QQQ dips, sells after recovery

Designed for uncertain markets: protects capital while capturing upside

github.com/verma-bipul/Algo-Trade-UMN-Team4